

LIOUVILLE AND LYAPUNOV CRITERIA FOR ANCIENT NAVIER–STOKES SOLUTIONS

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ABSTRACT. We study bounded ancient solutions of the three-dimensional Navier–Stokes equations arising from Type I blow-up arguments. We state standard PDE hypotheses under which such ancient solutions are forced to be spatially constant and hence vanish by the Type I bound. We also prove two Lyapunov rigidity results for rescaled vorticity equations.

For the unforced equations, known axisymmetric Liouville theorems cover the following cases: absence of swirl, pointwise scale-invariant control, finite $L_t^\infty L_x^p$ control of the swirl variable $\Gamma = ru_\theta$, and axial periodicity with bounded Γ . When one of these hypotheses holds for a nonzero Type I blow-up limit, the Galilean-trivial conclusions of the cited Liouville theorems reduce to the zero solution and contradict the nontriviality of the limit.

We also record two Lyapunov rigidity arguments. First, Gallay–Wayne’s small-data decay theorem for the three-dimensional rescaled vorticity equation yields a coercive Lyapunov functional forcing any ancient trajectory that stays in the perturbative weighted-vorticity ball to be zero. Second, Gallay–Maekawa’s stability theorem for the strained Burgers-vortex equation yields a Lyapunov functional for perturbations of a Burgers vortex and rigidity of ancient solutions in a small weighted neighborhood of the Burgers-vortex family.

Finally, for nonperiodic bounded-swirl axisymmetric solutions, we prove the exact scalar Γ -identity and the anisotropic cutoff-error decomposition. These identities give a coercive expanding-cutoff criterion that forces Γ to vanish, reducing the solution to the no-swirl case. The paper concludes with a summary of the Liouville hypotheses, conditional criteria, and cited PDE results.

1. INTRODUCTION

1.1. Ancient solutions from Type I blow-up. The Type I blow-up argument in Paper I [3] produces, after parabolic rescaling around a Type I singular point, a nonzero ancient solution (U, P) of

$$(1.1) \quad \partial_t U + (U \cdot \nabla)U + \nabla P = \Delta U, \quad \nabla \cdot U = 0, \quad t < 0,$$

satisfying the Type I ancient bound

$$(1.2) \quad \sup_{t < 0} \sqrt{-t} \|U(t)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Thus every result proving $U \equiv 0$ for an ancient solution satisfying (1.2) has the same consequence in the Type I problem:

$$\text{PDE hypothesis} \implies U \equiv 0 \implies \text{contradiction with nonzero blow-up limit.}$$

The present paper treats hypotheses for which a specific Liouville mechanism is available or for which a precise conditional criterion can be stated. The cases considered are:

axisymmetric Liouville hypotheses, small weighted rescaled vorticity,
the strained Burgers-vortex equation, bounded swirl with expanding cutoff control.

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No statement is made for decaying or symmetry-restricted ancient solutions outside the hypotheses listed below. Each conclusion is tied to an explicit PDE hypothesis and a specified analytic result.

In the Type I analysis of Paper I, we use the entries for which the stated PDE hypothesis and the cited or proved Liouville theorem imply $U \equiv 0$. Ancient profiles outside these hypotheses are not imported as structured exclusions in Paper I; they remain in the residual class and are handled by the residual closure paper [5].

1.2. Axisymmetric Liouville hypotheses. Several unforced axisymmetric cases are already covered by known Liouville theorems. Koch–Nadirashvili–Seregin–Šverák cover the no-swirl case and the pointwise scale-invariant condition $|u| \leq C/r$. The swirl-bearing finite- L^p condition

$$\Gamma = ru_\theta \in L_t^\infty L_x^p, \quad 1 \leq p < \infty,$$

is covered by Lei–Zhang–Zhao. The z -periodic case with bounded Γ is covered by Lei–Ren–Zhang.

The only additional step needed for Type I ancient limits is to convert Galilean-triviality into zero in the blow-up frame. We time-shift away from the possible singular time $t = 0$, apply the bounded ancient Liouville theorem, patch the resulting constants on backward intervals, and use (1.2): if $U(x, t) \equiv c$, then

$$\sup_{t < 0} \sqrt{-t} \|U(t)\|_\infty = \sup_{t < 0} \sqrt{-t} |c|$$

is finite only when $c = 0$.

1.3. Lyapunov hypotheses. The two Lyapunov results below use forward Lyapunov functionals. The first concerns the Gallay–Wayne perturbative weighted-vorticity regime. If w solves the rescaled three-dimensional vorticity equation and remains in the Gallay–Wayne small ball of $L_\sigma^2(m)$, $m > 7/2$, then the functional

$$(1.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2$$

is coercive and contracts along the forward semiflow. An ancient trajectory inside the small ball must therefore be zero, after sending the starting time to $-\infty$.

The second Lyapunov argument is for the strained Burgers-vortex equation

$$\partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (Ax \cdot \nabla) \Omega + A\Omega, \quad A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right),$$

not directly for the unforced Navier–Stokes equations. Gallay–Maekawa’s stability theorem gives a Lyapunov functional for perturbations with fixed circulation and forces ancient trajectories in a small weighted neighborhood of the Burgers-vortex family to be exactly stationary Burgers vortices. Applying that conclusion to an unforced Navier–Stokes solution requires the additional association assumption stated later.

1.4. The bounded-swirl case. The remaining axisymmetric case treated here is bounded swirl without finite- L^p decay or axial periodicity:

$$\Gamma = ru_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

For a smooth axisymmetric solution, the scalar Γ satisfies

$$(1.4) \quad \partial_t \Gamma + b \cdot \nabla \Gamma + \frac{2}{r} \partial_r \Gamma = \Delta \Gamma, \quad b = u_r e_r + u_z e_z.$$

The corresponding weighted identity is

$$(1.5) \quad \frac{1}{2} \frac{d}{dt} \int \Gamma^2 \phi \, dx + \int |\nabla \Gamma|^2 \phi \, dx = \frac{1}{2} \int \Gamma^2 \left(\partial_t \phi + \Delta \phi + b \cdot \nabla \phi + \frac{2}{r} \partial_r \phi \right) \, dx.$$

For anisotropic product cutoffs

$$\phi_{R,Z,\zeta}(r, z, t) = \lambda_R(r) \eta_Z(z - \zeta(t)),$$

the error density splits exactly into radial cutoff transport, axial drift mismatch, and cutoff curvature. The radial diffusion operator is $\partial_r^2 + 3r^{-1} \partial_r$, the radial Laplacian in four dimensions. We prove that a coercive expanding-cutoff hypothesis forces $\Gamma \equiv 0$. No assertion is made that every bounded nonperiodic Γ satisfies this additional hypothesis.

1.5. Main results. The first result records the unforced axisymmetric Liouville consequences relevant to the Type I argument.

Theorem 1.1 (Axisymmetric Liouville consequences for Type I ancient solutions). *Let U be a smooth Type I ancient solution of (1.1) satisfying (1.2). If U satisfies any of the following axisymmetric assumptions, then $U \equiv 0$:*

- (i) *axisymmetric without swirl;*
- (ii) *axisymmetric and satisfying a pointwise scale-invariant bound*

$$|U(x, t)| \leq \frac{C}{r};$$

- (iii) *axisymmetric with*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) *axisymmetric, periodic in the physical axial variable z , and with*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Consequently none of these assumptions can hold for a nonzero Type I blow-up limit.

Theorem 1.2 (Perturbative weighted-vorticity Lyapunov rigidity). *Let $m > \frac{7}{2}$. Let $w(\tau)$ be an ancient mild solution of the Galloway–Wayne rescaled vorticity equation*

$$(1.6) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla) w + (w \cdot \nabla) v, \quad \nabla \cdot w = 0,$$

with v recovered by the three-dimensional Biot–Savart law. Assume

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_{L_o^2(m)} \leq \rho_m,$$

where ρ_m is the Galloway–Wayne small-data radius. Then

$$w \equiv 0.$$

Here “ancient mild” means that every forward time-shift of w is the Galloway–Wayne mild solution launched from its value at the starting time.

Theorem 1.3 (Ancient rigidity near Burgers vortices). *Let $m > 2$. Let $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ be an admissible ancient mild solution of the strained Burgers-vortex vorticity equation*

$$(1.7) \quad \partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (A x \cdot \nabla) \Omega + A \Omega, \quad A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

Assume Ω remains in the Galloway–Maekawa small weighted neighborhood of the Burgers-vortex family, where $\bar{\alpha}$ denotes the circulation of Ω :

$$\sup_{\tau \in \mathbb{R}} \|\Omega(\tau) - \bar{\alpha} G\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha}, m}.$$

Then

$$\Omega(\tau) \equiv \bar{\alpha}G \quad \text{for all } \tau \in \mathbb{R}.$$

This is a result for the strained Burgers-vortex equation. It gives a statement about unforced Navier–Stokes ancient limits under the additional association assumption stated in [Theorem 7.7](#). The term “ancient mild” means compatibility with the forward Gallay–Maekawa semiflow after every time shift, as made explicit in [Theorem 7.6](#).

Theorem 1.4 (Localized scalar criterion for Γ). *Let u be a smooth bounded ancient mild axisymmetric solution of the unforced Navier–Stokes equations with bounded swirl variable*

$$\Gamma = ru_\theta \in L^\infty.$$

Assume the coercive expanding-cutoff hypotheses stated in [Theorem 8.5](#). Then

$$\Gamma \equiv 0 \quad \text{on } \mathbb{R}^3 \times (-\infty, 0).$$

Consequently, if the resulting no-swirl solution is covered by the standard axisymmetric no-swirl Liouville theorem, then u is Galilean-trivial. If, in addition, u is a Type I blow-up limit, then $u \equiv 0$.

Theorem 1.5 (Consequences for Type I blow-up limits). *Let U be a nonzero Type I blow-up limit. Then U cannot satisfy any of the axisymmetric assumptions in [Theorem 1.1](#), cannot have a corresponding rescaled vorticity satisfying the hypotheses of [Theorem 1.2](#), and cannot satisfy the bounded-swirl assumptions together with [Theorem 8.5](#). Independently, if a field Ω_U satisfies the assumptions in [Theorem 7.7](#), then $\Omega_U(\tau) \equiv \bar{\alpha}G$. This last conclusion is a rigidity statement for the strained Burgers-vortex equation and does not imply $U \equiv 0$ for the unforced Navier–Stokes equations.*

Only the explicit hypotheses stated above are used.

1.6. Relation to Papers I and II. Paper I [\[3\]](#) reduces Type I singularities to nonzero ancient limits satisfying [\(1.2\)](#). Paper II [\[4\]](#) treats uniformly L^3 -tight ancient solutions. The present paper treats the symmetry, decay, and Burgers-vortex hypotheses listed below. Known Liouville theorems rule out several cases, Lyapunov arguments rule out perturbative cases, and the remaining bounded-swirl axisymmetric case is reduced to a precise expanding-cutoff criterion. Paper III does not depend on Paper II.

1.7. Summary of hypotheses.

Hypothesis	Conclusion	Analytic ingredient
axisymmetric no swirl	$U \equiv 0$	KNSS
pointwise $ u \leq C/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$, $1 \leq p < \infty$	$U \equiv 0$	LZZ
z -periodic bounded Γ	$U \equiv 0$	LRZ
controlled swirl $\rho^\gamma V_\theta$	conditional	fixed-exponent Liouville hypothesis
weighted velocity decay	conditional	fixed-exponent Liouville hypothesis
small weighted vorticity	$w \equiv 0$	Gallay–Wayne
near Burgers vortices	$\Omega = \alpha G$	Gallay–Maekawa
association with Burgers vortices	conditional	Theorem 7.7
bounded nonperiodic $\Gamma \in L^\infty$	conditional	cutoff criterion needed
bounded Γ plus cutoff control	$\Gamma \equiv 0$	scalar identity plus no-swirl theorem

1.8. Organization. Section 2 records the Type I ancient setup, self-similar variables, the invariance facts, and the time-shift lemma. Section 3 fixes axisymmetric notation and the known Liouville theorems. Section 4 proves the unforced axisymmetric Liouville consequences. Section 5 records the controlled swirl and weighted velocity-decay conditions as conditional hypotheses. Section 6 proves the Gallay–Wayne Lyapunov rigidity. Section 7 proves rigidity for the Burgers-vortex equation. Section 8 derives the scalar Γ -identity, cutoff-error decomposition, and localized scalar theorem. Section 9 gives the final summary of consequences.

2. ANCIENT TYPE I LIMITS AND SELF-SIMILAR VARIABLES

2.1. Type I ancient solutions.

Definition 2.1 (Type I ancient solution). A smooth mild solution U of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$ is called Type I ancient if (1.2) holds. Mild is understood in the standard velocity formulation [10, 18]: for every $-\infty < s < t < 0$,

$$(2.1) \quad U(t) = e^{(t-s)\Delta}U(s) - \int_s^t e^{(t-\sigma)\Delta} \mathbb{P} \nabla \cdot (U(\sigma) \otimes U(\sigma)) d\sigma$$

in distributions, where $\mathbb{P}$ is the Leray projector.

2.2. Nonzero Type I blow-up limits.

Definition 2.2 (Nonzero Type I blow-up limit). A Type I ancient solution U is called a nonzero Type I blow-up limit if it arises as the limit of parabolic rescalings around a Type I singular point and the limiting solution is not identically zero. The arguments below require only the conclusion $U \not\equiv 0$.

Lemma 2.3 (Nontriviality of blow-up limits). *A nonzero Type I blow-up limit is not identically zero.*

Proof. This is part of the definition above. □

2.3. Backward self-similar variables.

For $t < 0$, define

$$(2.2) \quad V(y, \tau) = \sqrt{-t}U(x, t), \quad Q(y, \tau) = (-t)P(x, t), \quad y = \frac{x}{\sqrt{-t}}, \quad \tau = -\log(-t).$$

Equivalently,

$$x = e^{-\tau/2}y, \quad t = -e^{-\tau}.$$

Lemma 2.4 (Self-similar equation). *Let (U, P) be a smooth solution of (1.1), and define (V, Q) by (2.2). Then V is defined on $\mathbb{R}^3 \times \mathbb{R}$ and satisfies*

$$(2.3) \quad \partial_\tau V + (V \cdot \nabla)V + \nabla Q = \Delta V - \frac{1}{2}(V + y \cdot \nabla V), \quad \nabla \cdot V = 0.$$

Moreover (1.2) is equivalent to

$$\sup_{\tau \in \mathbb{R}} \|V(\tau)\|_{L^\infty(\mathbb{R}^3)} < \infty.$$

Proof. Let $a = \sqrt{-t} = e^{-\tau/2}$. Then $V = aU$, $y = x/a$, and

$$\partial_t \tau = \frac{1}{-t} = \frac{1}{a^2}, \quad \partial_t y = \frac{y}{2a^2}, \quad \partial_t(a^{-1}) = \frac{1}{2a^3}.$$

Since $U = a^{-1}V$, differentiating at fixed x gives

$$\partial_t U = a^{-3} \left(\partial_\tau V + \frac{1}{2}y \cdot \nabla V + \frac{1}{2}V \right).$$

Spatial derivatives scale as

$$\nabla_x U = a^{-2} \nabla_y V, \quad \Delta_x U = a^{-3} \Delta_y V,$$

and the pressure scaling gives

$$\nabla_x P = a^{-3} \nabla_y Q.$$

Thus

$$(U \cdot \nabla_x) U = a^{-3} (V \cdot \nabla_y) V.$$

Multiplying (1.1) by a^3 yields (2.3); the divergence equation follows from $\nabla_x \cdot U = a^{-2} \nabla_y \cdot V$. The spatial map $y \mapsto x = e^{-\tau/2} y$ is a bijection for each τ , and

$$\|V(\tau)\|_\infty = \sqrt{-t} \|U(t)\|_\infty.$$

Taking suprema over $t < 0$, equivalently $\tau \in \mathbb{R}$, gives the last assertion. $\square$

2.4. Preservation of axisymmetry and swirl variables. For axisymmetric flows we write

$$x = (r \cos \theta, r \sin \theta, z), \quad U = U_r e_r + U_\theta e_\theta + U_z e_z,$$

and in centered variables

$$y = (\rho \cos \theta, \rho \sin \theta, Z), \quad V = V_\rho e_\rho + V_\theta e_\theta + V_Z e_Z.$$

The scalar swirl variables are

$$\Gamma_U(r, z, t) = r U_\theta(r, z, t), \quad \Gamma_V(\rho, Z, \tau) = \rho V_\theta(\rho, Z, \tau).$$

Lemma 2.5 (Preservation of axisymmetric structure). *Axisymmetry and absence of swirl are preserved under (2.2). If U is axisymmetric, then*

$$\Gamma_V(\rho, Z, \tau) = \Gamma_U(r, z, t), \quad (r, z) = \sqrt{-t}(\rho, Z).$$

Fixed physical z -periodicity is preserved by physical time shifts, but under the centered change of variables it becomes a τ -dependent period in the Z -coordinate.

Proof. The self-similar map is a scalar spatial dilation and a scalar multiplication of the velocity, so it commutes with rotations around the symmetry axis. The cylindrical basis is unchanged by radial dilation, and

$$V_\theta(\rho, Z, \tau) = \sqrt{-t} U_\theta(r, z, t), \quad r = \sqrt{-t} \rho, \quad z = \sqrt{-t} Z.$$

Thus $U_\theta \equiv 0$ if and only if $V_\theta \equiv 0$, and

$$\Gamma_V = \rho V_\theta = \rho \sqrt{-t} U_\theta = r U_\theta = \Gamma_U.$$

If U has physical axial period L , then V has period $L/\sqrt{-t} = L e^{\tau/2}$ in Z , not a fixed period. $\square$

2.5. Time shifts away from the singular time.

Lemma 2.6 (Time shifts away from the singular time). *Let U be a smooth Type I ancient solution and fix $T < 0$. Then*

$$W_T(x, s) = U(x, T + s), \quad \Pi_T(x, s) = P(x, T + s), \quad s < 0,$$

is a smooth bounded mild ancient solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. Moreover the axisymmetric, no-swirl, L^p -swirl, bounded- Γ , and physical z -periodic properties are preserved.

Proof. Time translation preserves the differential equation because $\partial_s W_T(x, s) = \partial_t U(x, T + s)$ and all spatial derivatives are unchanged. For the mild formulation, apply (2.1) to U between the physical times $T + s < T + t < 0$:

$$U(T + t) = e^{(t-s)\Delta} U(T + s) - \int_{T+s}^{T+t} e^{(T+t-\eta)\Delta} \mathbb{P} \nabla \cdot (U(\eta) \otimes U(\eta)) d\eta.$$

With $\eta = T + \sigma$, this becomes the mild formula for W_T on the interval (s, t) . For $s < 0$, $T + s < T < 0$, so the Type I bound gives

$$\|W_T(s)\|_\infty = \|U(T + s)\|_\infty \leq \frac{1}{\sqrt{-T}} \sup_{\eta < 0} \sqrt{-\eta} \|U(\eta)\|_\infty.$$

The invariance assertions follow because only the time variable has been translated. In particular

$$\Gamma_{W_T}(r, z, s) = \Gamma_U(r, z, T + s),$$

so $L_s^\infty L_x^p$ and L^∞ bounds for Γ restrict to the shifted solution, and a fixed physical axial period remains the same. $\square$

2.6. PDE hypotheses used below. The arguments below are stated directly in terms of hypotheses on ancient solutions. The unconditional axisymmetric hypotheses are:

- (i) axisymmetry and $U_\theta \equiv 0$;
- (ii) axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$ for $r > 0$;
- (iii) axisymmetry and

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty;$$

- (iv) axisymmetry, periodicity in the physical axial variable z , and $\Gamma \in L^\infty$.

The additional hypotheses used later are the fixed-exponent controlled-swirl Liouville statement, the fixed-exponent weighted-velocity Liouville statement, the Gallay–Wayne small weighted-vorticity condition, the association with the Burgers-vortex equation, and the coercive expanding-cutoff condition [Theorem 8.5](#).

3. AXISYMMETRIC NOTATION AND KNOWN LIOUVILLE THEOREMS

3.1. Cylindrical variables and the swirl scalar. For an axisymmetric solution u , write

$$u = u_r e_r + u_\theta e_\theta + u_z e_z, \quad \Gamma = ru_\theta, \quad b = u_r e_r + u_z e_z.$$

When Γ is placed in $L^p(\mathbb{R}^3)$, it is regarded as the corresponding function of $x = (r \cos \theta, r \sin \theta, z)$, independent of θ :

$$\|\Gamma(t)\|_{L^p(\mathbb{R}^3)}^p = 2\pi \int_{\mathbb{R}} \int_0^\infty |\Gamma(r, z, t)|^p r dr dz, \quad 1 \leq p < \infty.$$

3.2. Constant and Galilean-trivial solutions.

Definition 3.1 (Galilean-trivial). A velocity field is Galilean-trivial if, after a Galilean transform, it is a constant vector field. In the Type I blow-up frame below we use only the following consequence: when a cited Liouville theorem gives a constant velocity field in that frame, the Type I bound forces the constant to be zero.

Lemma 3.2 (Spatially constant mild solutions). *Let W be a smooth mild solution of (1.1) on $\mathbb{R}^3 \times (-\infty, 0)$. If $W(x, t) = b(t)$ for all x and t , then b is constant in time.*

Proof. Insert $W(x, t) = b(t)$ into (2.1). The heat semigroup leaves constant vector fields fixed, and $\nabla \cdot (b(\sigma) \otimes b(\sigma)) = 0$. Thus $b(t) = b(s)$ for every $s < t < 0$. $\square$

Lemma 3.3 (Patching constants on backward intervals). *Let $U : \mathbb{R}^3 \times (-\infty, 0) \rightarrow \mathbb{R}^3$ be a function. Assume that for every $T < 0$ there exists $c_T \in \mathbb{R}^3$ such that*

$$U(x, t) = c_T \quad (x \in \mathbb{R}^3, t < T).$$

Then there is a single $c \in \mathbb{R}^3$ such that

$$U(x, t) = c \quad (x \in \mathbb{R}^3, t < 0).$$

Proof. If $T_1 < T_2 < 0$, the two constants agree on the nonempty interval $(-\infty, T_1)$. Hence all c_T are equal. For any $t < 0$, choose T with $t < T < 0$. $\square$

Lemma 3.4 (Constant Type I ancient solutions vanish). *Let U be a Type I ancient solution. If $U(x, t) \equiv c$, then $c = 0$.*

Proof. The Type I bound gives

$$\sqrt{-t} |c| \leq \sup_{s < 0} \sqrt{-s} \|U(s)\|_\infty < \infty \quad (t < 0).$$

Letting $t \rightarrow -\infty$ gives $c = 0$. $\square$

3.3. Known Liouville theorems. The following theorems are used as cited PDE results.

Theorem 3.5 (KNSS no-swirl Liouville theorem). *Every bounded ancient mild axisymmetric no-swirl solution of the unforced Navier–Stokes equations is Galilean-trivial. In the form used below, the solution is a spatially constant axial drift.*

Proof. This is the no-swirl Liouville theorem of Koch–Nadirashvili–Seregin–Šverák [11], stated in the form needed for the Type I vanishing argument. $\square$

Theorem 3.6 (KNSS pointwise scale-invariant condition). *Every bounded ancient mild axisymmetric solution satisfying*

$$|u(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0)$$

is Galilean-trivial. In the form used below it is identically zero.

Proof. This is the pointwise Liouville theorem of [11], stated in the form used here. $\square$

Theorem 3.7 (Lei–Zhang–Zhao). *Let u be a bounded ancient mild axisymmetric solution. If*

$$\Gamma = ru_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)), \quad 1 \leq p < \infty,$$

then u is a constant vector field.

Proof. This is the ancient axisymmetric L^p -swirl Liouville theorem of Lei–Zhang–Zhao [14]. $\square$

Theorem 3.8 (Lei–Ren–Zhang). *Let u be a bounded ancient mild axisymmetric solution, periodic in z , with*

$$\Gamma = ru_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then u is a constant axial vector field.

Proof. This is the periodic bounded-swirl theorem of Lei–Ren–Zhang [13]. Their statement is on $\mathbb{R}^2 \times \mathbb{T}^1$; a physical z -periodic whole-space solution descends to the quotient, and a Navier–Stokes scaling normalizes the period. $\square$

4. UNFORCED AXISYMMETRIC LIOUVILLE CONSEQUENCES

4.1. No swirl.

Proposition 4.1 (No-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and $U_\theta \equiv 0$. Then $U \equiv 0$.*

Proof. Fix $T < 0$ and set $W_T(x, s) = U(x, T + s)$. By Theorem 2.6, W_T is a bounded ancient mild axisymmetric no-swirl solution. Applying Theorem 3.5 gives that W_T is a spatially constant axial drift, say $W_T(x, s) = b_T(s)e_z$. By Theorem 3.2, $b_T(s)$ is constant in s . Thus $U(x, t) = c_T$ for $t < T$. Since $T < 0$ is arbitrary, Theorem 3.3 gives $U(x, t) = c$ for all $t < 0$. Finally, Theorem 3.4 gives $c = 0$. $\square$

4.2. Pointwise scale-invariant control.

Proposition 4.2 (Pointwise $1/r$ Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and satisfies*

$$|U(x, t)| \leq \frac{C}{r} \quad (r > 0, t < 0).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. The shifted solution $W_T(x, s) = U(x, T + s)$ is bounded, ancient, axisymmetric, and satisfies the same pointwise bound $|W_T| \leq C/r$. Theorem 3.6 gives $W_T \equiv 0$. Hence $U \equiv 0$ on $(-\infty, T)$. Since $T < 0$ is arbitrary, $U \equiv 0$. $\square$

 4.3. Finite L^p control of Γ .

Proposition 4.3 (Finite- L^p swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric and, for some $1 \leq p < \infty$,*

$$\Gamma = rU_\theta \in L_t^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, $W_T(x, s) = U(x, T + s)$ is a bounded ancient mild axisymmetric solution and

$$\Gamma_{W_T} \in L_s^\infty L_x^p(\mathbb{R}^3 \times (-\infty, 0)).$$

Theorem 3.7 gives that W_T is a constant vector field. Patch these constants in T by Theorem 3.3 and then use Theorem 3.4. $\square$

4.4. Periodic bounded swirl.

Proposition 4.4 (Periodic bounded-swirl Type I vanishing). *Let U be a smooth Type I ancient solution. Assume U is axisymmetric, periodic in the physical axial variable z , and satisfies*

$$\Gamma = rU_\theta \in L^\infty(\mathbb{R}^3 \times (-\infty, 0)).$$

Then $U \equiv 0$.

Proof. Fix $T < 0$. By Theorem 2.6, the shifted solution W_T is bounded, ancient, axisymmetric, physically z -periodic, and has bounded Γ . Theorem 3.8 gives that W_T is a constant axial vector field. The constants patch as T varies, and the Type I bound eliminates the resulting constant by Theorem 3.4. $\square$

4.5. Proof of the axisymmetric Liouville theorem.

Proof of Theorem 1.1. The four cases are exactly the hypotheses of Theorems 4.1 to 4.4. Each proposition gives $U \equiv 0$. This contradicts Theorem 2.3 when U is a nonzero Type I blow-up limit. $\square$

5. CONDITIONAL CONTROLLED-SWIRL AND WEIGHTED-DECAY HYPOTHESES

5.1. Controlled swirl in self-similar variables.

Definition 5.1 (Controlled-swirl condition). Fix $\gamma > 0$. A centered ancient solution V satisfies the controlled-swirl condition with exponent γ if V is axisymmetric and

$$\sup_{\tau \in \mathbb{R}} \sup_{\rho > 0, Z \in \mathbb{R}} \rho^\gamma |V_\theta(\rho, Z, \tau)| < \infty.$$

Hypothesis 5.2 (Fixed-exponent controlled-swirl Liouville theorem). For the chosen exponent $\gamma > 0$, every Type I ancient solution whose centered representative satisfies the controlled-swirl condition with exponent γ is identically zero.

Proposition 5.3 (Conditional controlled-swirl vanishing). *If Theorem 5.2 holds for a fixed $\gamma > 0$, then no nonzero Type I blow-up limit can have a centered representative satisfying the controlled-swirl condition with exponent γ .*

Proof. This is exactly the hypothesis combined with the nontriviality Theorem 2.3. $\square$

5.2. Weighted velocity decay.

Definition 5.4 (Weighted velocity-decay condition). Fix $m > 0$. A centered ancient solution V satisfies the weighted velocity-decay condition with exponent m if

$$\sup_{\tau \in \mathbb{R}} \int_{\mathbb{R}^3} (1 + |y|^2)^m |V(y, \tau)|^2 dy < \infty.$$

Hypothesis 5.5 (Fixed-exponent weighted velocity Liouville theorem). For the chosen exponent $m > 0$, every Type I ancient solution whose centered representative satisfies the weighted velocity-decay condition with exponent m is identically zero.

Proposition 5.6 (Conditional weighted velocity vanishing). *If Theorem 5.5 holds for a fixed $m > 0$, then no nonzero Type I blow-up limit can have a centered representative satisfying the weighted velocity-decay condition with exponent m .*

Proof. This is Theorem 5.5 plus Theorem 2.3. $\square$

Remark 5.7 (Weighted velocity versus weighted vorticity). The weighted velocity condition is not covered by the Gallay–Wayne theorem below. Gallay–Wayne applies only to ancient trajectories whose rescaled vorticity remains in a sufficiently small ball of $L_\sigma^2(m)$. Thus Theorem 5.5 remains an independent fixed-exponent Liouville hypothesis unless weighted velocity decay is shown to imply the corresponding perturbative weighted-vorticity condition.

6. PERTURBATIVE WEIGHTED-VORTICITY LYAPUNOV RIGIDITY

6.1. The rescaled vorticity equation. In this section τ is the forward self-similar time used in the Gallay–Wayne vorticity formulation. The ancient trajectories are indexed by all $\tau \in \mathbb{R}$, and the forward semiflow acts by increasing τ .

Let $w = \nabla \times v$. The rescaled vorticity equation is

$$(6.1) \quad \partial_\tau w = \Lambda w - (v \cdot \nabla)w + (w \cdot \nabla)v, \quad \nabla \cdot w = 0,$$

where

$$(6.2) \quad \Lambda w = \Delta w + \frac{1}{2} \xi \cdot \nabla w + w,$$

and v is recovered from w by the three-dimensional Biot–Savart law. For $m \geq 0$, set

$$L_\sigma^2(m) = \{f \in L^2(\mathbb{R}^3; \mathbb{R}^3) : \nabla \cdot f = 0, \quad \|f\|_m < \infty\},$$

where

$$\|f\|_m^2 = \int_{\mathbb{R}^3} (1 + |\xi|)^{2m} |f(\xi)|^2 d\xi.$$

6.2. Galloway–Wayne small-data decay.

Theorem 6.1 (Galloway–Wayne small-data decay). *Let $0 < \mu \leq 1$ and $m > 2\mu + \frac{1}{2}$. There exist constants $\rho_m > 0$ and $C_m \geq 1$ such that if $\|w_0\|_m \leq \rho_m$, then (6.1) has a unique global mild solution $w(\tau) = \Phi_\tau w_0$ in $C([0, \infty); L_\sigma^2(m))$ and*

$$\|\Phi_\tau w_0\|_m \leq C_m e^{-\mu\tau} \|w_0\|_m, \quad \tau \geq 0.$$

In particular, if $m > \frac{7}{2}$, one may use $\mu = 1$.

Proof. This is the Galloway–Wayne weighted small-data theorem for the rescaled three-dimensional vorticity equation [7]. The only conclusions used below are existence, uniqueness, the semiflow property, and the displayed exponential decay. $\square$

6.3. The Lyapunov functional.

Definition 6.2 (Perturbative weighted-vorticity condition). Fix $m > \frac{7}{2}$, and let ρ_m be the radius in Theorem 6.1. An ancient vorticity trajectory satisfies the perturbative weighted-vorticity condition with exponent m if it is a function

$$w \in C(\mathbb{R}; L_\sigma^2(m))$$

that solves (6.1) for all $\tau \in \mathbb{R}$ and satisfies

$$\sup_{\tau \in \mathbb{R}} \|w(\tau)\|_m \leq \rho_m.$$

Solving for all τ means that every time shift is compatible with the Galloway–Wayne forward mild semiflow.

Proposition 6.3 (Semiflow Lyapunov functional). *Fix $m > \frac{7}{2}$. For $\|w_0\|_m \leq \rho_m$, define*

$$(6.3) \quad \mathcal{L}_m(w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_\tau w_0\|_m^2.$$

Then $\mathcal{L}_m$ is well-defined and satisfies

$$(6.4) \quad \|w_0\|_m^2 \leq \mathcal{L}_m(w_0) \leq C_m^2 \|w_0\|_m^2.$$

Moreover, for every $s \geq 0$,

$$(6.5) \quad \mathcal{L}_m(\Phi_s w_0) \leq e^{-2s} \mathcal{L}_m(w_0).$$

Proof. The Galloway–Wayne estimate with $\mu = 1$ gives

$$e^{2\tau} \|\Phi_\tau w_0\|_m^2 \leq C_m^2 \|w_0\|_m^2,$$

which proves finiteness and the upper bound. The lower bound is the value at $\tau = 0$. By uniqueness,

$$\Phi_\tau(\Phi_s w_0) = \Phi_{\tau+s} w_0.$$

Hence

$$\mathcal{L}_m(\Phi_s w_0) = \sup_{\tau \geq 0} e^{2\tau} \|\Phi_{\tau+s} w_0\|_m^2 = e^{-2s} \sup_{\sigma \geq s} e^{2\sigma} \|\Phi_\sigma w_0\|_m^2 \leq e^{-2s} \mathcal{L}_m(w_0).$$

□

6.4. Ancient rigidity in the small weighted ball.

Proof of Theorem 1.2. Let w be an ancient solution in the small ball and fix $\tau_0 \in \mathbb{R}$. For $s < \tau_0$, the shifted function

$$W_s(\theta) = w(s + \theta), \quad \theta \geq 0,$$

is the Gally–Wayne mild solution with initial datum $w(s)$. The small-ball hypothesis gives $\|w(s)\|_m \leq \rho_m$, so the forward solution is in the domain of Φ_θ , and uniqueness gives

$$W_s(\theta) = \Phi_\theta w(s), \quad \theta \geq 0.$$

Choosing $\theta = \tau_0 - s$ gives

$$w(\tau_0) = \Phi_{\tau_0-s} w(s).$$

Since $\|w(\tau_0)\|_m \leq \rho_m$, the Lyapunov functional is also defined at $w(\tau_0)$. Applying Theorem 6.3 with forward time $\tau_0 - s$,

$$\mathcal{L}_m(w(\tau_0)) \leq e^{-2(\tau_0-s)} \mathcal{L}_m(w(s)) \leq C_m^2 \rho_m^2 e^{-2(\tau_0-s)}.$$

Letting $s \rightarrow -\infty$ gives $\mathcal{L}_m(w(\tau_0)) = 0$. The lower coercive bound (6.4) gives $w(\tau_0) = 0$. Since τ_0 was arbitrary, $w \equiv 0$. □

Remark 6.4 (Relation to weighted velocity decay). This theorem applies only to Type I blow-up limits whose vorticity satisfies the Gally–Wayne perturbative weighted condition. It does not exclude all weighted velocity-decay conditions.

7. THE BURGERS-VORTEX EQUATION

This section proves a rigidity theorem for the strained Burgers-vortex equation. It is not a theorem for the unforced Navier–Stokes equations. The final corollary states the additional association assumption needed to transfer the conclusion to an unforced ancient-solution class.

7.1. The strained vorticity equation. Let $x = (x_h, x_3) \in \mathbb{R}^2 \times \mathbb{R}$ and

$$A = \text{diag} \left(-\frac{1}{2}, -\frac{1}{2}, 1 \right).$$

The Burgers-vortex vorticity equation is

$$(7.1) \quad \partial_\tau \Omega + (U \cdot \nabla) \Omega - (\Omega \cdot \nabla) U = \Delta \Omega - (Ax \cdot \nabla) \Omega + A \Omega, \quad \nabla \cdot \Omega = 0,$$

with U recovered from Ω by the Biot–Savart law. It has the stationary family

$$(7.2) \quad \Omega = \alpha G, \quad G(x) = \begin{pmatrix} 0 \\ 0 \\ g(x_h) \end{pmatrix}, \quad g(x_h) = \frac{1}{4\pi} e^{-|x_h|^2/4}, \quad \alpha \in \mathbb{R}.$$

7.2. Weighted horizontal spaces. For $m > 1$, let $L_h^2(m)$ denote the horizontal weighted space with norm

$$\|f\|_{L_h^2(m)}^2 = \int_{\mathbb{R}^2} |f(x_h)|^2 \left(1 + \frac{|x_h|^2}{4m}\right)^m dx_h.$$

Set

$$X(m) = BC(\mathbb{R}_{x_3}; L_h^2(m)), \quad X_0(m) = \left\{ f \in X(m) : \int_{\mathbb{R}^2} f(x_h, x_3) dx_h = 0 \text{ for every } x_3 \right\},$$

and

$$\mathcal{X}_{\text{full}}(m) = X(m)^3, \quad \mathcal{X}_{\text{mod}}(m) = X(m) \times X(m) \times X_0(m).$$

7.3. Admissible Burgers-vortex solutions.

Definition 7.1 (Admissible Burgers-vortex solutions). A mild solution $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ is called admissible if the following operations are justified on every compact time interval:

- (i) horizontal integrations by parts against constants and against x_h are valid, with no boundary contribution at $|x_h| = \infty$;
- (ii) the identity

$$\int_{\mathbb{R}^2} [(U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U]_3 dx_h = 0$$

holds in distributions in x_3 and τ ;

- (iii) the map

$$\tau \mapsto \int_{\mathbb{R}^2} \Omega_3(x_h, x_3, \tau) dx_h$$

is absolutely continuous after testing in x_3 .

Smooth rapidly decaying solutions are admissible.

7.4. Circulation coordinate and conservation.

Lemma 7.2 (Circulation coordinate). *Let $m > 1$ and let $\Omega \in \mathcal{X}_{\text{full}}(m)$ be divergence-free. Assume that the horizontal integration by parts*

$$\int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h(x_h, x_3) dx_h = 0$$

is valid. Then

$$\alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \Omega_3(x_h, x_3) dx_h$$

is well-defined and independent of x_3 . Writing the common value as $\alpha[\Omega]$, one has

$$Q\Omega := \Omega - \alpha[\Omega]G \in \mathcal{X}_{\text{mod}}(m).$$

Proof. Since $m > 1$, $L_h^2(m) \hookrightarrow L^1(\mathbb{R}^2)$ by Cauchy-Schwarz:

$$\int_{\mathbb{R}^2} |f(x_h)| dx_h \leq \|f\|_{L_h^2(m)} \left(\int_{\mathbb{R}^2} \left(1 + \frac{|x_h|^2}{4m}\right)^{-m} dx_h \right)^{1/2},$$

and the last integral is finite in two dimensions exactly because $m > 1$. Thus $\alpha[\Omega](x_3)$ is well-defined. Using $\nabla \cdot \Omega = 0$,

$$\partial_{x_3} \alpha[\Omega](x_3) = \int_{\mathbb{R}^2} \partial_{x_3} \Omega_3 dx_h = - \int_{\mathbb{R}^2} \nabla_h \cdot \Omega_h dx_h = 0.$$

Also $\int_{\mathbb{R}^2} g(x_h) dx_h = 1$, so subtracting $\alpha[\Omega]G$ removes the horizontal mean of the third component. $\square$

Lemma 7.3 (Conservation of circulation). *Let $m > 1$, and let $\Omega(\tau)$ be an admissible mild solution of (7.1) with values in $\mathcal{X}_{\text{full}}(m)$. Then $\alpha[\Omega(\tau)]$ is independent of τ . If $\bar{\alpha}$ denotes this constant, then*

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

belongs to $\mathcal{X}_{\text{mod}}(m)$ for every τ and solves the fixed-circulation perturbation equation around $\bar{\alpha}G$.

Proof. Integrate the third component of (7.1) over x_h . The nonlinear term is a horizontal divergence:

$$(U \cdot \nabla)\Omega - (\Omega \cdot \nabla)U = -\nabla \times (U \times \Omega),$$

and the third component of a curl is a horizontal divergence. It therefore has zero horizontal integral by admissibility. For the linear part,

$$[\Delta\Omega - (Ax \cdot \nabla)\Omega + A\Omega]_3 = \Delta_h \Omega_3 + \partial_{x_3}^2 \Omega_3 + \frac{1}{2}x_h \cdot \nabla_h \Omega_3 - x_3 \partial_{x_3} \Omega_3 + \Omega_3.$$

The Δ_h term integrates to zero, and

$$\int_{\mathbb{R}^2} \frac{1}{2}x_h \cdot \nabla_h \Omega_3 dx_h = - \int_{\mathbb{R}^2} \Omega_3 dx_h,$$

which cancels the $+\Omega_3$ term. Hence

$$\frac{d}{d\tau} \alpha[\Omega(\tau)] = \partial_{x_3}^2 \alpha[\Omega(\tau)] - x_3 \partial_{x_3} \alpha[\Omega(\tau)].$$

By Theorem 7.2, the right-hand side is zero. $\square$

7.5. Galloway–Maekawa stability.

Theorem 7.4 (Galloway–Maekawa stability). *Let $m > 2$ and $\alpha \in \mathbb{R}$. There exist constants $\delta_{\alpha,m} > 0$ and $C_{\alpha,m} \geq 1$ such that every divergence-free perturbation*

$$\omega_0 \in \mathcal{X}_{\text{mod}}(m), \quad \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\alpha,m},$$

generates a unique global mild solution of the fixed-circulation perturbation equation around αG , denoted $\Phi_\tau^\alpha \omega_0$, and

$$\|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq C_{\alpha,m} e^{-\tau/2} \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \quad (\tau \geq 0).$$

Proof. This is the Galloway–Maekawa nonlinear stability theorem for Burgers vortices in horizontal weighted spaces [9]. We use only existence, uniqueness, the fixed-circulation semiflow property, and the displayed exponential decay. $\square$

7.6. Lyapunov functional near a Burgers vortex.

Proposition 7.5 (Lyapunov functional relative to a Burgers vortex). *Fix $m > 2$ and $\alpha \in \mathbb{R}$. On the ball*

$$\|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\alpha,m}$$

define

$$(7.3) \quad \mathcal{B}_{\alpha,m}(\omega_0) = \sup_{\tau \geq 0} e^\tau \|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

Then

$$\|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq \mathcal{B}_{\alpha,m}(\omega_0) \leq C_{\alpha,m}^2 \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2,$$

and

$$\mathcal{B}_{\alpha,m}(\Phi_s^\alpha \omega_0) \leq e^{-s} \mathcal{B}_{\alpha,m}(\omega_0) \quad (s \geq 0).$$

Proof. Gally–Maekawa stability gives, for every $\tau \geq 0$,

$$e^\tau \|\Phi_\tau^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq C_{\alpha,m}^2 \|\omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

Taking the supremum over $\tau \geq 0$ proves that $\mathcal{B}_{\alpha,m}$ is finite and gives the upper coercive bound. The lower bound follows from the value at $\tau = 0$, because $\Phi_0^\alpha \omega_0 = \omega_0$.

For $s \geq 0$, uniqueness of the fixed-circulation perturbation equation gives the semiflow identity

$$\Phi_\tau^\alpha(\Phi_s^\alpha \omega_0) = \Phi_{\tau+s}^\alpha \omega_0.$$

Therefore

$$\mathcal{B}_{\alpha,m}(\Phi_s^\alpha \omega_0) = \sup_{\tau \geq 0} e^\tau \|\Phi_{\tau+s}^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2.$$

With $\sigma = \tau + s$, this is

$$e^{-s} \sup_{\sigma \geq s} e^\sigma \|\Phi_\sigma^\alpha \omega_0\|_{\mathcal{X}_{\text{mod}}(m)}^2 \leq e^{-s} \mathcal{B}_{\alpha,m}(\omega_0).$$

□

7.7. Ancient rigidity near the Burgers-vortex family.

Definition 7.6 (Small ancient solutions near Burgers vortices). Fix $m > 2$. We say that an admissible ancient mild solution $\Omega : \mathbb{R} \rightarrow \mathcal{X}_{\text{full}}(m)$ of (7.1) stays in the Gally–Maekawa neighborhood of the Burgers-vortex family if, with $\bar{\alpha} = \alpha[\Omega(\tau)]$,

$$\sup_{\tau \in \mathbb{R}} \|\Omega(\tau) - \bar{\alpha}G\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha},m}.$$

The term “ancient mild” includes compatibility with the Gally–Maekawa forward mild semiflow after every time shift.

Proof of Theorem 1.3. By Theorem 7.3, $\bar{\alpha}$ is independent of time and

$$\omega(\tau) = \Omega(\tau) - \bar{\alpha}G$$

is an ancient solution of the fixed-circulation perturbation equation in the small ball. Fix $\tau_0 \in \mathbb{R}$ and $s < \tau_0$. By the ancient mild compatibility, the shifted perturbation

$$\omega_s(\theta) = \omega(s + \theta), \quad \theta \geq 0,$$

is the Gally–Maekawa mild solution launched from $\omega(s)$. The neighborhood hypothesis gives

$$\|\omega(s)\|_{\mathcal{X}_{\text{mod}}(m)} \leq \delta_{\bar{\alpha},m},$$

so uniqueness of the fixed-circulation semiflow gives

$$\omega(\tau_0) = \Phi_{\tau_0-s}^{\bar{\alpha}} \omega(s).$$

Using Theorem 7.5,

$$\mathcal{B}_{\bar{\alpha},m}(\omega(\tau_0)) \leq e^{-(\tau_0-s)} \mathcal{B}_{\bar{\alpha},m}(\omega(s)) \leq e^{-(\tau_0-s)} C_{\bar{\alpha},m}^2 \delta_{\bar{\alpha},m}^2.$$

Letting $s \rightarrow -\infty$ gives $\mathcal{B}_{\bar{\alpha},m}(\omega(\tau_0)) = 0$. The coercive lower bound gives $\omega(\tau_0) = 0$. Since τ_0 was arbitrary, $\Omega(\tau) \equiv \bar{\alpha}G$. □

7.8. Conditional association with unforced Navier–Stokes.

Assumption 7.7 (Association with the Burgers-vortex equation). Let $\mathcal{C}$ be a class of ancient solutions of an unforced or renormalized Navier–Stokes problem. Assume that $m > 2$ and that each $u \in \mathcal{C}$ is equipped with a field

$$\Omega_u \in C_{\text{loc}}(\mathbb{R}; \mathcal{X}_{\text{full}}(m))$$

and a nonnegative function $\varepsilon_u(\tau)$ such that:

- (i) Ω_u is an admissible ancient mild solution of (7.1) in the sense of Theorem 7.1;
- (ii) with $\bar{\alpha} = \alpha[\Omega_u]$,

$$\|\Omega_u(\tau) - \bar{\alpha}G\|_{\mathcal{X}_{\text{mod}}(m)} \leq C_* \varepsilon_u(\tau) \quad (\tau \in \mathbb{R});$$

- (iii) $\sup_{\tau \in \mathbb{R}} \varepsilon_u(\tau) \leq \delta_{\bar{\alpha}, m}/C_*$.

Corollary 7.8 (Conditional Burgers-vortex conclusion). Under Theorem 7.7, the corresponding field for every $u \in \mathcal{C}$ is an exact Burgers vortex:

$$\Omega_u(\tau) \equiv \bar{\alpha}G.$$

Proof. The assumptions place Ω_u in the small Galloway–Maekawa neighborhood defined in Theorem 7.6. Applying Theorem 1.3 gives the conclusion. $\square$

8. THE SCALAR Γ IDENTITY AND CUTOFF CRITERION

8.1. The equation for Γ .

Lemma 8.1 (Swirl equation). Let u be a smooth axisymmetric solution of (1.1). Then $\Gamma = ru_\theta$ satisfies

$$(8.1) \quad \partial_t \Gamma + b \cdot \nabla \Gamma + \frac{2}{r} \partial_r \Gamma = \Delta \Gamma, \quad b = u_r e_r + u_z e_z.$$

Proof. The angular component of (1.1) is

$$\partial_t u_\theta + u_r \partial_r u_\theta + u_z \partial_z u_\theta + \frac{u_r}{r} u_\theta = \left(\partial_r^2 + \frac{1}{r} \partial_r + \partial_z^2 - \frac{1}{r^2} \right) u_\theta.$$

Multiplying by r , the transport terms become

$$ru_r \partial_r u_\theta + u_r u_\theta = u_r \partial_r (ru_\theta) = u_r \partial_r \Gamma, \quad ru_z \partial_z u_\theta = u_z \partial_z \Gamma.$$

For the radial diffusion, use $u_\theta = \Gamma/r$. Then

$$\partial_r u_\theta = \frac{1}{r} \partial_r \Gamma - \frac{1}{r^2} \Gamma, \quad \partial_r^2 u_\theta = \frac{1}{r} \partial_r^2 \Gamma - \frac{2}{r^2} \partial_r \Gamma + \frac{2}{r^3} \Gamma.$$

Consequently

$$r \left(\partial_r^2 + \frac{1}{r} \partial_r - \frac{1}{r^2} \right) u_\theta = \partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma.$$

The axial diffusion satisfies $r \partial_z^2 u_\theta = \partial_z^2 \Gamma$. Since

$$\Delta \Gamma = \partial_r^2 \Gamma + \frac{1}{r} \partial_r \Gamma + \partial_z^2 \Gamma,$$

the total diffusion term after multiplication by r is

$$\partial_r^2 \Gamma - \frac{1}{r} \partial_r \Gamma + \partial_z^2 \Gamma = \Delta \Gamma - \frac{2}{r} \partial_r \Gamma.$$

Substitution gives

$$\partial_t \Gamma + b \cdot \nabla \Gamma = \Delta \Gamma - \frac{2}{r} \partial_r \Gamma,$$

which is (8.1). □

8.2. The weighted scalar identity.

Proposition 8.2 (Weighted scalar identity). *Let u be a smooth axisymmetric solution of (1.1). Set $W = \Gamma$, and let $\phi = \phi(r, z, t) \geq 0$ be a smooth compactly supported axisymmetric weight. Then*

$$(8.2) \quad \frac{1}{2} \frac{d}{dt} \int_{\mathbb{R}^3} W^2 \phi \, dx + \int_{\mathbb{R}^3} |\nabla W|^2 \phi \, dx = \frac{1}{2} \int_{\mathbb{R}^3} W^2 \left(\partial_t \phi + \Delta \phi + b \cdot \nabla \phi + \frac{2}{r} \partial_r \phi \right) \, dx.$$

If $W = \Gamma - c$, the same identity holds provided $c = 0$ or ϕ vanishes near the axis. Without this restriction, the right-hand side must be augmented by the axis boundary term

$$2\pi c^2 \int_{\mathbb{R}} \phi(0, z, t) \, dz.$$

Proof. From Theorem 8.1,

$$\partial_t W = \Delta W - b \cdot \nabla W - \frac{2}{r} \partial_r W.$$

Multiply by $W\phi$ and integrate. The time derivative gives

$$\int W \partial_t W \phi \, dx = \frac{1}{2} \frac{d}{dt} \int W^2 \phi \, dx - \frac{1}{2} \int W^2 \partial_t \phi \, dx.$$

The Laplacian gives

$$\int W \Delta W \phi \, dx = - \int |\nabla W|^2 \phi \, dx + \frac{1}{2} \int W^2 \Delta \phi \, dx.$$

Because u is divergence-free and b is its meridional part, $\nabla \cdot b = 0$ in the three-dimensional sense:

$$\nabla \cdot b = \frac{1}{r} \partial_r(r u_r) + \partial_z u_z = \nabla \cdot u = 0.$$

Thus, using compact support of ϕ ,

$$- \int W(b \cdot \nabla W) \phi \, dx = \frac{1}{2} \int W^2 b \cdot \nabla \phi \, dx.$$

Finally, in cylindrical variables $dx = r \, dr \, d\theta \, dz$,

$$- \int W \frac{2}{r} \partial_r W \phi \, dx = - \int_0^{2\pi} \int_{\mathbb{R}} \int_0^\infty \partial_r(W^2) \phi \, dr \, dz \, d\theta.$$

For $W = \Gamma$, smoothness gives $\Gamma = O(r^2)$ at the axis, so the boundary term at $r = 0$ vanishes. More explicitly, after integrating on $\delta < r < \infty$ and letting $\delta \downarrow 0$, the boundary contribution is

$$\int_0^{2\pi} \int_{\mathbb{R}} W(\delta, z, t)^2 \phi(\delta, z, t) \, dz \, d\theta \longrightarrow 0.$$

Therefore the last display equals

$$\int W^2 \frac{1}{r} \partial_r \phi \, dx = \frac{1}{2} \int W^2 \frac{2}{r} \partial_r \phi \, dx.$$

Combining the terms proves (8.2). If $W = \Gamma - c$, then $W(0, z, t) = -c$. The same integration on $\delta < r < \infty$ leaves the limit

$$\int_0^{2\pi} \int_{\mathbb{R}} c^2 \phi(0, z, t) \, dz \, d\theta = 2\pi c^2 \int_{\mathbb{R}} \phi(0, z, t) \, dz,$$

which is absent exactly when $c = 0$ or ϕ avoids the axis. □

8.3. Anisotropic product cutoffs.

$$\phi_{R,Z,\zeta}(r,z,t) = \lambda_R(r)\eta_Z(z - \zeta(t)),$$

where λ_R, η_Z are smooth cutoffs and ζ is a smooth axial center.

Proposition 8.3 (Cutoff-error decomposition). *Let $W = \Gamma$. With the notation above,*

$$(8.3) \quad \frac{1}{2} \frac{d}{dt} \int W^2 \lambda_R \eta_Z dx + \int |\nabla W|^2 \lambda_R \eta_Z dx = \frac{1}{2} \int W^2 \mathcal{E}_{R,Z,\zeta}[u] dx,$$

where

$$(8.4) \quad \mathcal{E}_{R,Z,\zeta}[u] = \eta_Z \left(\lambda_R'' + \frac{3}{r} \lambda_R' + u_r \lambda_R' \right) + \lambda_R \left(\eta_Z'' + (u_z - \dot{\zeta}) \eta_Z' \right).$$

Proof. For an axisymmetric scalar,

$$\Delta(\lambda_R \eta_Z) = \left(\lambda_R'' + \frac{1}{r} \lambda_R' \right) \eta_Z + \lambda_R \eta_Z''.$$

Also

$$\frac{2}{r} \partial_r (\lambda_R \eta_Z) = \frac{2}{r} \lambda_R' \eta_Z, \quad b \cdot \nabla (\lambda_R \eta_Z) = u_r \lambda_R' \eta_Z + u_z \lambda_R \eta_Z',$$

and

$$\partial_t (\lambda_R \eta_Z) = -\dot{\zeta} \lambda_R \eta_Z'.$$

Adding these four contributions gives (8.4); substitution into [Theorem 8.2](#) gives (8.3). $\square$

Remark 8.4 (The radial geometry). The differential expression

$$\lambda_R'' + \frac{3}{r} \lambda_R'$$

is the radial Laplacian in four dimensions. Thus the natural radial barrier for the Γ equation is adapted to $\partial_r^2 + 3r^{-1} \partial_r$. The cutoff errors in (8.4) are precisely radial transport, axial drift mismatch, and the two cutoff-curvature terms.

8.4. Coercive expanding-cutoff control.

Assumption 8.5 (Coercive expanding-cutoff control). Let u be a bounded ancient mild axisymmetric solution with bounded Γ . Assume in addition that u is smooth and that the identities [Theorems 8.1](#) to [8.3](#) hold for the weights below. There exist expanding smooth weights

$$\phi_n(r, z, t) = \lambda_n(r) \eta_n(z - \zeta_n(t)), \quad 0 \leq \phi_n \leq 1, \quad \phi_n \uparrow 1$$

locally uniformly on compact subsets of $\mathbb{R}^3 \times (-\infty, 0)$, compactly supported in space for each fixed time. There are constants $a_n > 0$, $M_n < \infty$, and $\varepsilon_n \geq 0$, with $\varepsilon_n/a_n \rightarrow 0$, such that

$$L_n(t) := \int_{\mathbb{R}^3} \Gamma^2 \phi_n(t) dx < \infty$$

for every $t < 0$, L_n is locally absolutely continuous, and, for each n , the following estimates hold for a.e. $\sigma < 0$:

$$(8.5) \quad \int_{\mathbb{R}^3} |\nabla \Gamma|^2 \phi_n(\sigma) dx \geq a_n L_n(\sigma),$$

$$(8.6) \quad \left| \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u] dx \right| \leq 2\varepsilon_n,$$

and

$$(8.7) \quad \sup_{\sigma < t} L_n(\sigma) \leq M_n.$$

Here $\mathcal{E}_n[u]$ is the cutoff-error density associated with ϕ_n .

8.5. Scalar conclusion from expanding cutoffs.

Proof of Theorem 1.4. Fix n . Integrating (8.3) from s to t , with $-\infty < s < t < 0$, gives

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + \int_s^t \int_{\mathbb{R}^3} |\nabla \Gamma|^2 \phi_n \, dx \, d\sigma = \frac{1}{2} \int_s^t \int_{\mathbb{R}^3} \Gamma^2 \mathcal{E}_n[u] \, dx \, d\sigma.$$

Using (8.5) and (8.6), this implies

$$\frac{1}{2}L_n(t) - \frac{1}{2}L_n(s) + a_n \int_s^t L_n(\sigma) \, d\sigma \leq \varepsilon_n(t - s).$$

Since L_n is locally absolutely continuous, the preceding integral inequality yields the differential inequality

$$\frac{d}{dt}L_n(t) + 2a_n L_n(t) \leq 2\varepsilon_n$$

for a.e. $t < 0$. Multiplying by $e^{2a_n t}$ and integrating from s to t gives

$$L_n(t) \leq e^{-2a_n(t-s)} L_n(s) + \frac{\varepsilon_n}{a_n} \left(1 - e^{-2a_n(t-s)}\right).$$

Using the past bound (8.7) and letting $s \rightarrow -\infty$ yields

$$L_n(t) \leq \frac{\varepsilon_n}{a_n}.$$

Let $K \subset \mathbb{R}^3$ be compact and fix $t < 0$. Since $\phi_n \uparrow 1$ locally uniformly in space-time, for every $\eta \in (0, 1)$ there is $n_0 = n_0(K, t, \eta)$ such that

$$\phi_n(x, t) \geq 1 - \eta \quad (x \in K, n \geq n_0).$$

Therefore, for $n \geq n_0$,

$$(1 - \eta) \int_K \Gamma(x, t)^2 \, dx \leq L_n(t) \leq \frac{\varepsilon_n}{a_n}.$$

Letting $n \rightarrow \infty$ and using $\varepsilon_n/a_n \rightarrow 0$ gives

$$\int_K \Gamma(x, t)^2 \, dx = 0$$

for every compact K and every $t < 0$. Since Γ is continuous by the smoothness in Theorem 8.5, local L^2 -vanishing on every compact set implies $\Gamma(x, t) = 0$ pointwise for every x and $t < 0$. Thus $\Gamma \equiv 0$. $\square$

8.6. Reduction after scalar vanishing.

Corollary 8.6 (Reduction after scalar vanishing). *Let u be as in Theorem 1.4. If in addition u is a nonzero Type I blow-up limit in the same physical frame, then $\Gamma \equiv 0$, $u_\theta \equiv 0$, and $u \equiv 0$, contradicting the nontriviality of the blow-up limit.*

Proof. Theorem 1.4 gives $\Gamma \equiv 0$. Then $u_\theta = 0$ for $r > 0$, so the no-swirl Type I vanishing result Theorem 4.1 applies. $\square$

9. SUMMARY OF PDE CONSEQUENCES

9.1. Zero conclusions.

Theorem 9.1 (Zero conclusions for Type I blow-up limits). *The following hypotheses cannot hold for a nonzero Type I blow-up limit:*

- (i) *axisymmetry and $U_\theta \equiv 0$;*
- (ii) *axisymmetry and the pointwise bound $|U(x, t)| \leq C/r$;*
- (iii) *axisymmetry and finite- L^p control of the swirl scalar,*

$$\Gamma \in L_t^\infty L_x^p, \quad 1 \leq p < \infty;$$

- (iv) *axisymmetry, z -periodicity, and bounded Γ ;*
- (v) *the Galloway–Wayne small weighted-vorticity condition, whenever the velocity is recovered from the corresponding vorticity by the Biot–Savart law;*
- (vi) *bounded Γ together with the expanding-cutoff hypotheses in [Theorem 8.5](#).*

Proof. Items (i)–(iv) are [Theorem 1.1](#). Item (v) is [Theorem 1.2](#); since the velocity in that case is recovered from w by the Biot–Savart law, $w \equiv 0$ gives the zero velocity. Item (vi) is [Theorem 8.6](#). In each Type I case, the zero conclusion contradicts [Theorem 2.3](#). $\square$

9.2. Burgers-vortex rigidity.

Theorem 9.2 (Burgers-vortex rigidity). *Every admissible ancient solution of the strained Burgers-vortex equation that stays in the Galloway–Maekawa neighborhood of the Burgers-vortex family is a stationary Burgers vortex αG . Under the association assumption [Theorem 7.7](#), the same conclusion holds for the corresponding field Ω_u .*

Proof. The rigidity statement is [Theorem 1.3](#); the conditional association statement is [Theorem 7.8](#). $\square$

9.3. Conditional Liouville hypotheses.

Proposition 9.3 (Conditional Liouville hypotheses). *Under [Theorem 5.2](#), no nonzero Type I blow-up limit can satisfy the controlled-swirl condition with exponent γ . Under [Theorem 5.5](#), no nonzero Type I blow-up limit can satisfy the weighted velocity-decay condition with exponent m .*

Proof. This is [Theorems 5.3](#) and [5.6](#). $\square$

9.4. Bounded swirl without cutoff control.

Remark 9.4 (Nonperiodic bounded swirl without the cutoff criterion). The full class of bounded ancient axisymmetric unforced Navier–Stokes solutions with nontrivial bounded Γ , no finite- L^p decay of Γ , no axial periodicity, and no verified expanding-cutoff estimate remains outside the unconditional conclusions here. The identity [\(8.3\)](#) reduces this case to explicit transport and cutoff terms, but it does not eliminate those terms without the additional estimates in [Theorem 8.5](#).

9.5. Consequences for Type I blow-up arguments.

Proof of [Theorem 1.5](#). The axisymmetric cases are ruled out by [Theorem 1.1](#). The perturbative weighted-vorticity case is ruled out by [Theorem 1.2](#) once the corresponding rescaled vorticity satisfies the small weighted condition; the Biot–Savart reconstruction then gives zero velocity. The bounded-swirl case under [Theorem 8.5](#) is ruled out by [Theorem 8.6](#). These zero conclusions contradict the nontriviality of a Type I blow-up limit. The Burgers-vortex conclusion is separate: under [Theorem 7.7](#), [Theorem 7.8](#) gives $\Omega_U(\tau) \equiv \bar{\alpha}G$. That strained-equation theorem alone asserts no zero-solution conclusion for the unforced equations. $\square$

Theorem 9.5 (Combined zero conclusion). *Let U be a Type I ancient solution obtained as a nonzero Type I blow-up limit. If U satisfies one of the hypotheses in [Theorem 9.1](#), or one of the conditional hypotheses in [Theorem 9.3](#) for which the stated Liouville theorem is available, then $U \equiv 0$, a contradiction. The Burgers-vortex equation contributes only the separate rigidity statement of [Theorem 9.2](#).*

Proof. This is [Theorem 9.1](#) and [Theorem 9.3](#), with the Burgers-vortex assertion recorded separately in [Theorem 9.2](#). $\square$

9.6. Summary table.

Hypothesis	Conclusion	Analytic ingredient
no swirl	$U \equiv 0$	KNSS plus Type I constant removal
pointwise $1/r$	$U \equiv 0$	KNSS
$\Gamma \in L_t^\infty L_x^p$	$U \equiv 0$	LZZ plus Type I constant removal
periodic bounded Γ	$U \equiv 0$	LRZ plus Type I constant removal
controlled swirl $\rho^\gamma V_\theta$	conditional	Theorem 5.2
weighted velocity decay	conditional	Theorem 5.5
small weighted vorticity	$w \equiv 0$	Gallay–Wayne Lyapunov functional
near Burgers vortices	$\Omega = \alpha G$	Gallay–Maekawa Lyapunov functional
association with Burgers vortices	conditional	Theorem 7.7
bounded nonperiodic $\Gamma \in L^\infty$	conditional	Theorem 8.5 needed
bounded Γ plus cutoff control	$\Gamma \equiv 0$	scalar identity plus no-swirl theorem

The no-swirl, pointwise scale-invariant, finite- L^p -swirl, and periodic bounded-swirl cases are covered by known axisymmetric Liouville theorems. The perturbative weighted-vorticity case is covered by a Lyapunov functional derived from Gallay–Wayne decay, and small ancient perturbations of Burgers vortices are rigid for the strained equation by Gallay–Maekawa stability. The remaining nonperiodic bounded-swirl case is reduced here to the coercive expanding-cutoff criterion for the scalar $\Gamma = ru_\theta$. Thus every zero conclusion used in the Type I application is attached to an explicit PDE hypothesis stated in this paper.

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